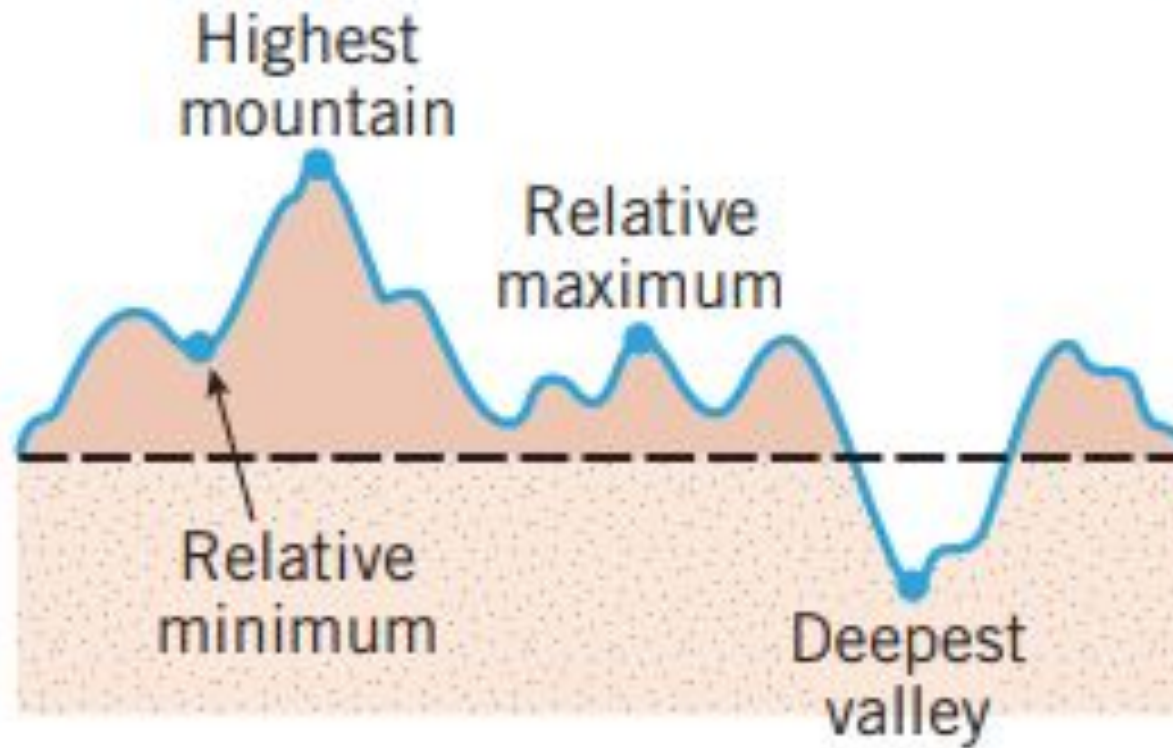


MAXIMA AND MINIMA , OPTIMIZATION

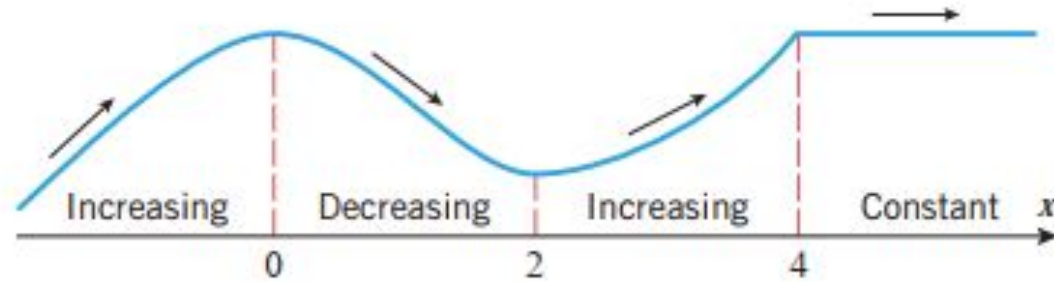
RELATIVE MAXIMA AND MINIMA



▲ Figure 4.2.1



Theory

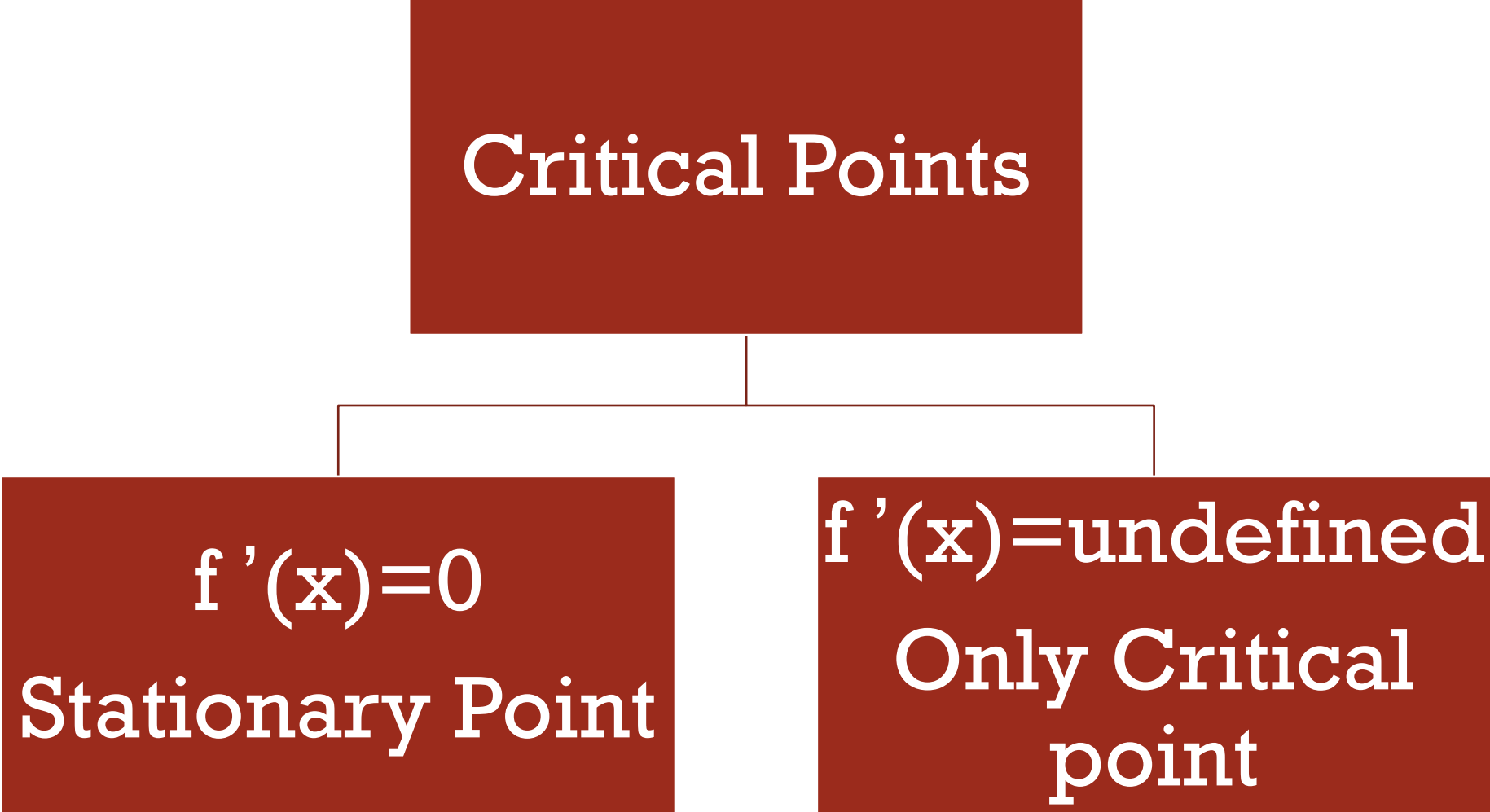


4.1.2 THEOREM Let f be a function that is continuous on a closed interval $[a, b]$ and differentiable on the open interval (a, b) .

- (a) If $f'(x) > 0$ for every value of x in (a, b) , then f is increasing on $[a, b]$.
- (b) If $f'(x) < 0$ for every value of x in (a, b) , then f is decreasing on $[a, b]$.
- (c) If $f'(x) = 0$ for every value of x in (a, b) , then f is constant on $[a, b]$.



Critical Points



```
graph TD; A[Critical Points] --> B["f'(x)=0<br/>Stationary Point"]; A --> C["f'(x)=undefined<br/>Only Critical point"]
```

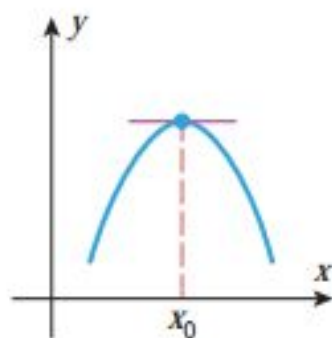
$$f'(x)=0$$

Stationary Point

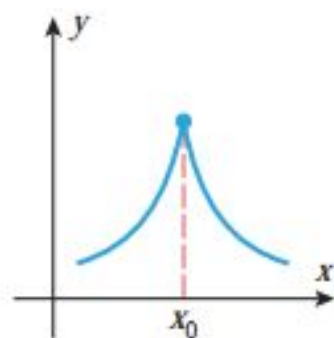
$$f'(x)=\text{undefined}$$

Only Critical
point

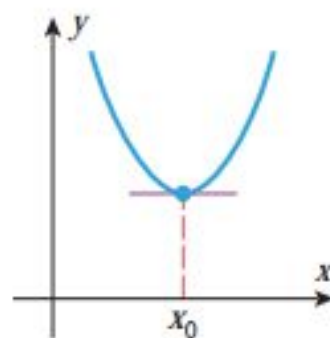




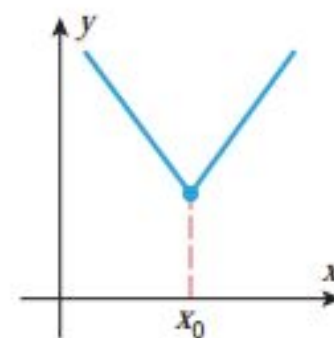
Critical point
Stationary point
Relative maximum



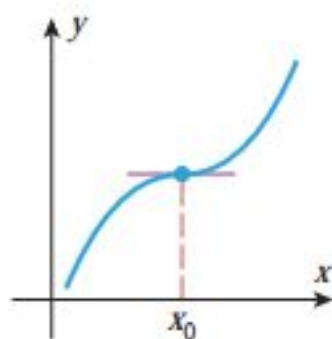
Critical point
Not a stationary point
Relative maximum



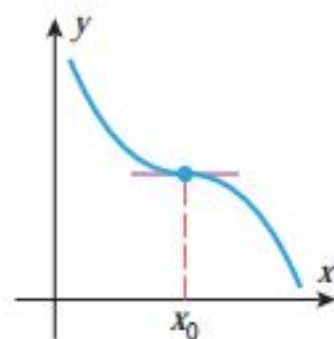
Critical point
Stationary point
Relative minimum



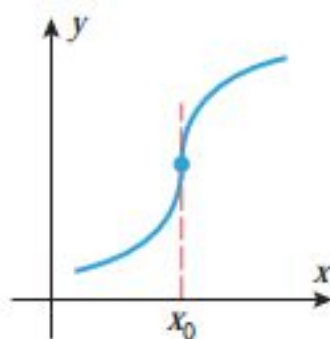
Critical point
Not a stationary point
Relative minimum



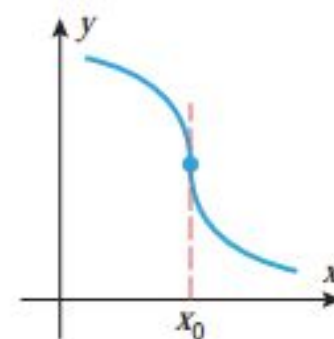
Critical point
Stationary point
Inflection point
Not a relative extremum



Critical point
Stationary point
Inflection point
Not a relative extremum



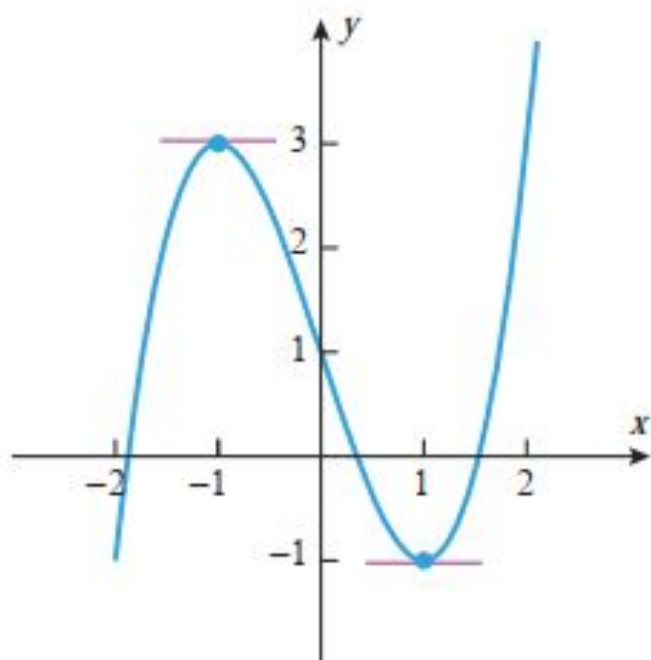
Critical point
Not a stationary point
Inflection point
Not a relative extremum



Critical point
Not a stationary point
Inflection point
Not a relative extremum

▲ Figure 4.2.6





$$y = x^3 - 3x + 1$$

▲ Figure 4.2.4

4.2.2 THEOREM Suppose that f is a function defined on an open interval containing the point x_0 . If f has a relative extremum at $x = x_0$, then $x = x_0$ is a critical point of f ; that is, either $f'(x_0) = 0$ or f is not differentiable at x_0 .

► **Example 2** Find all critical points of $f(x) = x^3 - 3x + 1$.

Solution. The function f , being a polynomial, is differentiable everywhere, so its critical points are all stationary points. To find these points we must solve the equation $f'(x) = 0$. Since

$$f'(x) = 3x^2 - 3 = 3(x + 1)(x - 1)$$

we conclude that the critical points occur at $x = -1$ and $x = 1$. This is consistent with the graph of f in Figure 4.2.4. ◀



► **Example 5** Figure 4.1.10 shows the graph of the function $f(x) = x^3 - 3x^2 + 1$. Use the first and second derivatives of f to determine the intervals on which f is increasing, decreasing, concave up, and concave down. Locate all inflection points and confirm that your conclusions are consistent with the graph.

Solution. Calculating the first two derivatives of f we obtain

$$f'(x) = 3x^2 - 6x = 3x(x - 2)$$

$$f''(x) = 6x - 6 = 6(x - 1)$$

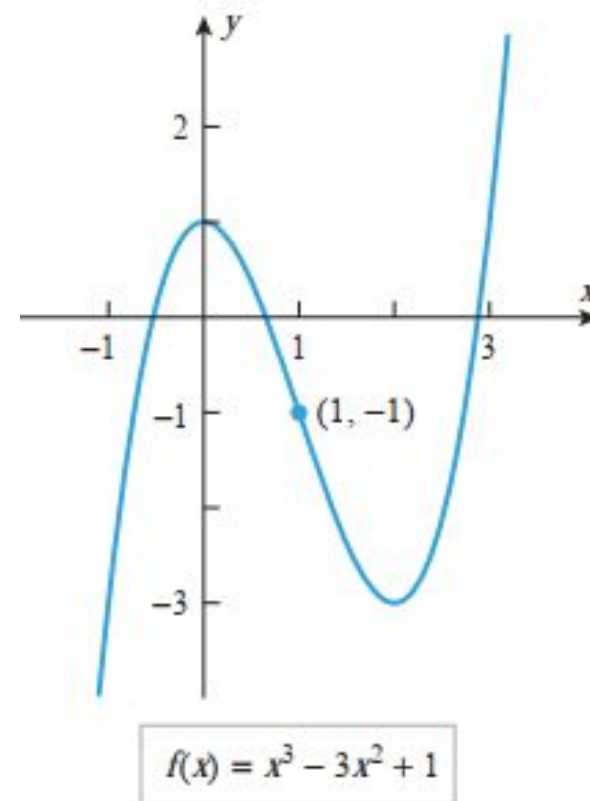
The sign analysis of these derivatives is shown in the following tables:

INTERVAL	$(3x)(x-2)$	$f'(x)$	CONCLUSION
$x < 0$	$(-)(-)$	$+$	f is increasing on $(-\infty, 0]$
$0 < x < 2$	$(+)(-)$	$-$	f is decreasing on $[0, 2]$
$x > 2$	$(+)(+)$	$+$	f is increasing on $[2, +\infty)$

INTERVAL	$6(x-1)$	$f''(x)$	CONCLUSION
$x < 1$	$(-)$	$-$	f is concave down on $(-\infty, 1)$
$x > 1$	$(+)$	$+$	f is concave up on $(1, +\infty)$

The second table shows that there is an inflection point at $x = 1$, since f changes from concave down to concave up at that point. The inflection point is $(1, f(1)) = (1, -1)$. All of these conclusions are consistent with the graph of f . ◀

Answer should be in open brackets



► **Example 6** Figure 4.1.11 suggests that the function $f(x) = xe^{-x}$ has an inflection point but its exact location is not evident from the graph in this figure. Use the first and second derivatives of f to determine the intervals on which f is increasing, decreasing, concave up, and concave down. Locate all inflection points.

Solution. Calculating the first two derivatives of f we obtain (verify)

$$f'(x) = (1 - x)e^{-x}$$

$$f''(x) = (x - 2)e^{-x}$$

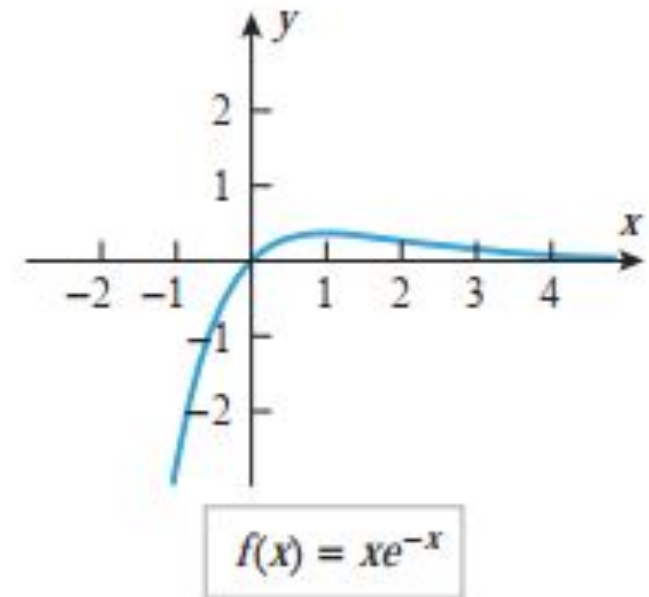
Keeping in mind that e^{-x} is positive for all x , the sign analysis of these derivatives is easily determined:

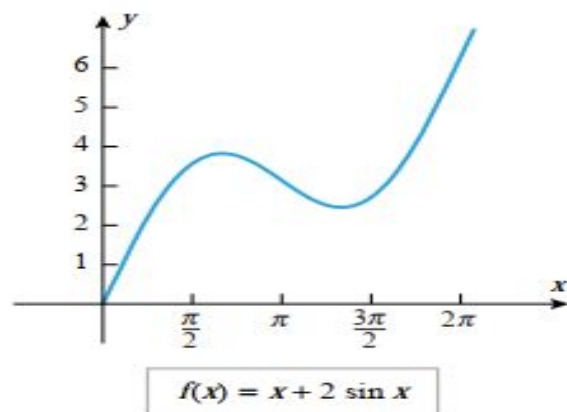
INTERVAL	$(1 - x)(e^{-x})$	$f'(x)$	CONCLUSION
$x < 1$	$(+)(+)$	$+$	f is increasing on $(-\infty, 1]$
$x > 1$	$(-)(+)$	$-$	f is decreasing on $[1, +\infty)$

INTERVAL	$(x - 2)(e^{-x})$	$f''(x)$	CONCLUSION
$x < 2$	$(-)(+)$	$-$	f is concave down on $(-\infty, 2)$
$x > 2$	$(+)(+)$	$+$	f is concave up on $(2, +\infty)$

The second table shows that there is an inflection point at $x = 2$, since f changes from concave down to concave up at that point. All of these conclusions are consistent with the graph of f . ◀

$$\begin{aligned} f'(x) &= (1-x)e^{-x} \\ \text{Let } f'(x) &= 0 \\ 0 &= (1-x)e^{-x} \\ (1-x) &= 0, \text{ or } e^{-x} = 0 \\ x &= 1, \text{ does not exist} \end{aligned} \quad \left| \quad \begin{aligned} f''(x) &= -e^{-x} + (1-x)(-e^{-x}) \\ &= -e^{-x} - e^{-x} + e^{-x}x \\ &= (x-2)e^{-x} \\ \text{Let } f''(x) &= 0 \\ 0 &= (x-2)e^{-x} \\ x &= 2, \end{aligned}$$





► Figure 4.1.12

► **Example 7** Figure 4.1.12 shows the graph of the function $f(x) = x + 2 \sin x$ over the interval $[0, 2\pi]$. Use the first and second derivatives of f to determine where f is increasing, decreasing, concave up, and concave down. Locate all inflection points and confirm that your conclusions are consistent with the graph.

Solution. Calculating the first two derivatives of f we obtain

$$f'(x) = 1 + 2 \cos x$$

$$f''(x) = -2 \sin x$$

Since f' is a continuous function, it changes sign on the interval $(0, 2\pi)$ only at points where $f'(x) = 0$ (why?). These values are solutions of the equation

$$1 + 2 \cos x = 0 \quad \text{or equivalently} \quad \cos x = -\frac{1}{2}$$

There are two solutions of this equation in the interval $(0, 2\pi)$, namely, $x = 2\pi/3$ and $x = 4\pi/3$ (verify). Similarly, f'' is a continuous function, so its sign changes in the interval $(0, 2\pi)$ will occur only at values of x for which $f''(x) = 0$. These values are solutions of the equation

$$-2 \sin x = 0$$

There is one solution of this equation in the interval $(0, 2\pi)$, namely, $x = \pi$. With the help of these “sign transition points” we obtain the sign analysis shown in the following tables:

INTERVAL	$f'(x) = 1 + 2 \cos x$	CONCLUSION
$0 < x < 2\pi/3$	+	f is increasing on $[0, 2\pi/3]$
$2\pi/3 < x < 4\pi/3$	−	f is decreasing on $[2\pi/3, 4\pi/3]$
$4\pi/3 < x < 2\pi$	+	f is increasing on $[4\pi/3, 2\pi]$

INTERVAL	$f''(x) = -2 \sin x$	CONCLUSION
$0 < x < \pi$	−	f is concave down on $(0, \pi)$
$\pi < x < 2\pi$	+	f is concave up on $(\pi, 2\pi)$

The second table shows that there is an inflection point at $x = \pi$, since f changes from concave down to concave up at that point. All of these conclusions are consistent with the graph of f . ◀

Answer should be in open brackets



► **Example 3** Find all critical points of $f(x) = 3x^{5/3} - 15x^{2/3}$.

Solution. The function f is continuous everywhere and its derivative is

$$f'(x) = 5x^{2/3} - 10x^{-1/3} = 5x^{-1/3}(x - 2) = \frac{5(x - 2)}{x^{1/3}}$$

We see from this that $f'(x) = 0$ if $x = 2$ and $f'(x)$ is undefined if $x = 0$. Thus $x = 0$ and $x = 2$ are critical points and $x = 2$ is a stationary point. This is consistent with the graph of f shown in Figure 4.2.5. ◀

Table 4.2.1

INTERVAL	$5(x - 2)/x^{1/3}$	$f'(x)$
$x < 0$	$(-)/(-)$	$+$
$0 < x < 2$	$(-)/(+)$	$-$
$x > 2$	$(+)/(+)$	$+$



► **Example 6** Find $\lim_{x \rightarrow 0} (1 + \sin x)^{1/x}$.

Solution. As discussed above, we begin by introducing a dependent variable

$$y = (1 + \sin x)^{1/x}$$

and taking the natural logarithm of both sides:

$$\ln y = \ln(1 + \sin x)^{1/x} = \frac{1}{x} \ln(1 + \sin x) = \frac{\ln(1 + \sin x)}{x}$$

Thus,

$$\lim_{x \rightarrow 0} \ln y = \lim_{x \rightarrow 0} \frac{\ln(1 + \sin x)}{x}$$

which is an indeterminate form of type $0/0$, so by L'Hôpital's rule

$$\lim_{x \rightarrow 0} \ln y = \lim_{x \rightarrow 0} \frac{\ln(1 + \sin x)}{x} = \lim_{x \rightarrow 0} \frac{(\cos x)/(1 + \sin x)}{1} = 1$$

Since we have shown that $\ln y \rightarrow 1$ as $x \rightarrow 0$, the continuity of the exponential function implies that $e^{\ln y} \rightarrow e^1$ as $x \rightarrow 0$, and this implies that $y \rightarrow e$ as $x \rightarrow 0$. Thus,

$$\lim_{x \rightarrow 0} (1 + \sin x)^{1/x} = e \quad \blacktriangleleft$$



$$f(x) = x^4 - 8x^2 + 16$$

$$f'(x) = 4x^3 - 16x$$

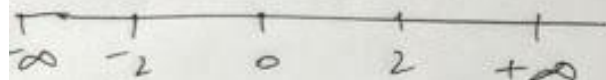
$$\text{Let } f'(x) = 0$$

$$0 = 4x^3 - 16x$$

$$4x(x^2 - 4) = 0$$

$$\Rightarrow 4x(x+2)(x-2) = 0$$

$$x = 0, x = -2, x = 2$$



Interval	$4x(x+2)(x-2)$	$f'(x)$	Conclusion
$x < -2$	$(-)(-)(-)$	$(-)$	$f(x)$ is decreasing on $(-\infty, -2)$
$-2 < x < 0$	$(-)(+)(-)$	$(+)$	$f(x)$ is increasing on $(-2, 0)$
$0 < x < 2$	$(+)(+)(-)$	$(-)$	$f(x)$ is decreasing on $(0, 2)$
$x > 2$	$(+)(+)(+)$	$(+)$	$f(x)$ is increasing on $(2, +\infty)$

$$f''(x) = 12x^2 - 16 = 4(3x^2 - 4) = 4(\sqrt{3}x + 2)(\sqrt{3}x - 2)$$

Interval	$4(3x^2 - 4)$	$f''(x)$	Conclusion
$x < -\frac{2}{\sqrt{3}}$	$(+)(+)$	$(+)$	$f(x)$ is concave up $(-\infty, -\frac{2}{\sqrt{3}})$
$-\frac{2}{\sqrt{3}} < x < \frac{2}{\sqrt{3}}$	$(+)(-)$	$(-)$	$f(x)$ is concave down $(-\frac{2}{\sqrt{3}}, \frac{2}{\sqrt{3}})$
$x > \frac{2}{\sqrt{3}}$	$(+)(+)$	$(+)$	$f(x)$ is concave up $(\frac{2}{\sqrt{3}}, +\infty)$

$x = -\frac{2}{\sqrt{3}}$ & $x = \frac{2}{\sqrt{3}}$ are inflection points.



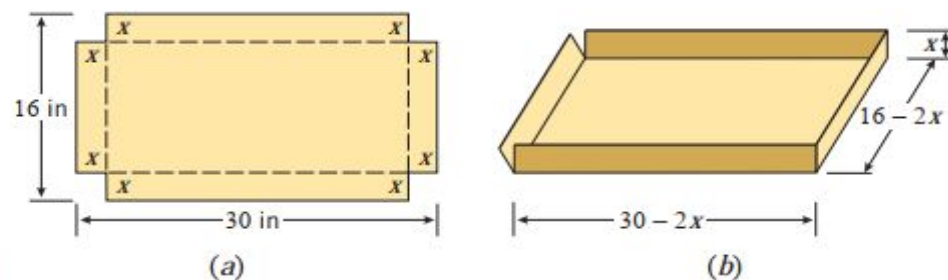
The Derivative in Graphing and Applications (pg.279)

A Procedure for Solving Applied Maximum and Minimum Problems

- Step 1.** Draw an appropriate figure and label the quantities relevant to the problem.
- Step 2.** Find a formula for the quantity to be maximized or minimized.
- Step 3.** Using the conditions stated in the problem to eliminate variables, express the quantity to be maximized or minimized as a function of one variable.
- Step 4.** Find the interval of possible values for this variable from the physical restrictions in the problem.
- Step 5.** If applicable, use the techniques of the preceding section to obtain the maximum or minimum.



► **Example 2** An open box is to be made from a 16-inch by 30-inch piece of cardboard by cutting out squares of equal size from the four corners and bending up the sides (Figure 4.5.3). What size should the squares be to obtain a box with the largest volume?



► Figure 4.5.3

Solution. For emphasis, we explicitly list the steps of the five-step problem-solving procedure given above as an outline for the solution of this problem. (In later examples we will follow these guidelines without listing the steps.)

- **Step 1:** Figure 4.5.3a illustrates the cardboard piece with squares removed from its corners. Let

x = length (in inches) of the sides of the squares to be cut out

V = volume (in cubic inches) of the resulting box

- **Step 2:** Because we are removing a square of side x from each corner, the resulting box will have dimensions $16 - 2x$ by $30 - 2x$ by x (Figure 4.5.3b). Since the volume of a box is the product of its dimensions, we have

$$V = (16 - 2x)(30 - 2x)x = 480x - 92x^2 + 4x^3 \quad (5)$$

- **Step 3:** Note that our volume expression is already in terms of the single variable x .
- **Step 4:** The variable x in (5) is subject to certain restrictions. Because x represents a length, it cannot be negative, and because the width of the cardboard is 16 inches, we cannot cut out squares whose sides are more than 8 inches long. Thus, the variable x in (5) must satisfy

$$0 \leq x \leq 8$$

and hence we have reduced our problem to finding the value (or values) of x in the interval $[0, 8]$ for which (5) is a maximum.

- **Step 5:** From (5) we obtain

$$\begin{aligned} \frac{dV}{dx} &= 480 - 184x + 12x^2 = 4(120 - 46x + 3x^2) \\ &= 4(x - 12)(3x - 10) \end{aligned}$$

Setting $dV/dx = 0$ yields

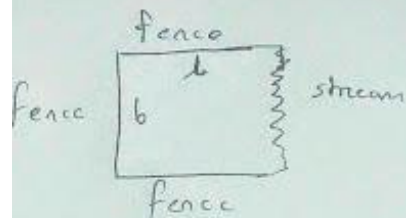
$$x = \frac{10}{3} \quad \text{and} \quad x = 12$$

Since $x = 12$ falls outside the interval $[0, 8]$, the maximum value of V occurs either at the critical point $x = \frac{10}{3}$ or at the endpoints $x = 0$, $x = 8$. Substituting these values into (5) yields Table 4.5.2, which tells us that the greatest possible volume $V = \frac{19,600}{27} \text{ in}^3 \approx 726 \text{ in}^3$ occurs when we cut out squares whose sides have length $\frac{10}{3}$ inches. This is consistent with the graph of (5) shown in Figure 4.5.4. ◀

Table 4.5.2

x	0	$\frac{10}{3}$	8
V	0	$\frac{19,600}{27} \approx 726$	0





Perimeter of ~~fencing~~ fence $\Rightarrow 2l + b = 1000 - \text{cst}$

~~Perimeter~~ $b = 1000 - 2l$

$$\text{Area} = l \times b = l(1000 - 2l)$$

$$A = 1000l - 2l^2$$

$$\frac{dA}{dl} = 1000 - 4l$$

At extreme points $\frac{dA}{dl} = 0$

$$0 = 1000 - 4l$$

$$\Rightarrow l = 250 \text{ ~~ft~~ ft}$$

$\therefore$ To determine nature of value we would use the 2nd derivative test.
derivative test.

$$\frac{d^2A}{dl^2} = -4, \text{ so the } l=250 \text{ provide maximum value}$$

$$A = 1000(250) - 2(250)^2$$

$$= 125,000 \text{ ft}^2$$

1ST DERIVATIVE TEST

4.2.3 THEOREM (*First Derivative Test*) Suppose that f is continuous at a critical point x_0 .

- (a) If $f'(x) > 0$ on an open interval extending left from x_0 and $f'(x) < 0$ on an open interval extending right from x_0 , then f has a relative maximum at x_0 .
- (b) If $f'(x) < 0$ on an open interval extending left from x_0 and $f'(x) > 0$ on an open interval extending right from x_0 , then f has a relative minimum at x_0 .
- (c) If $f'(x)$ has the same sign on an open interval extending left from x_0 as it does on an open interval extending right from x_0 , then f does not have a relative extremum at x_0 .



$$f(x) = 3x^{5/3} - 15x^{2/3}$$

$$f'(x) = 3 \times \frac{5}{3} x^{5/3-1} - 15 \times \frac{2}{3} x^{2/3-1}$$

$$= 5x^{2/3} - 10x^{-1/3}$$

$$= 5x^{2/3} - \frac{10}{x^{1/3}}$$

$$= \frac{5x^{2/3}x^{1/3} - 10}{x^{1/3}} = \frac{5x - 10}{x^{1/3}} = \frac{5(x-2)}{x^{1/3}}$$

$$\text{Let } f'(x) = 0, \quad 0 = \frac{5(x-2)}{x^{1/3}}$$

$$\Rightarrow 5(x-2) = 0$$

$$\Rightarrow x = 2$$

$$\text{If } x=0, \quad f'(0) = \frac{5(0-2)}{0} = \text{undefined}$$

Thus, $x=0$ and $x=2$ are critical points and $x=2$ is a stationary point

$$f'(x) = \frac{5(x-2)}{x^{1/3}}$$

interval	$5(x-2)/x^{1/3}$	$f'(x)$
$x < 0$	$(-)/(-)$	$(+)$, increasing
$0 < x < 2$	$(-)/(+)$	$(-)$, decreasing
$x > 2$	$(+)/(+)$	$(+)$, increasing

sign changes (between $x < 0$ and $0 < x < 2$)

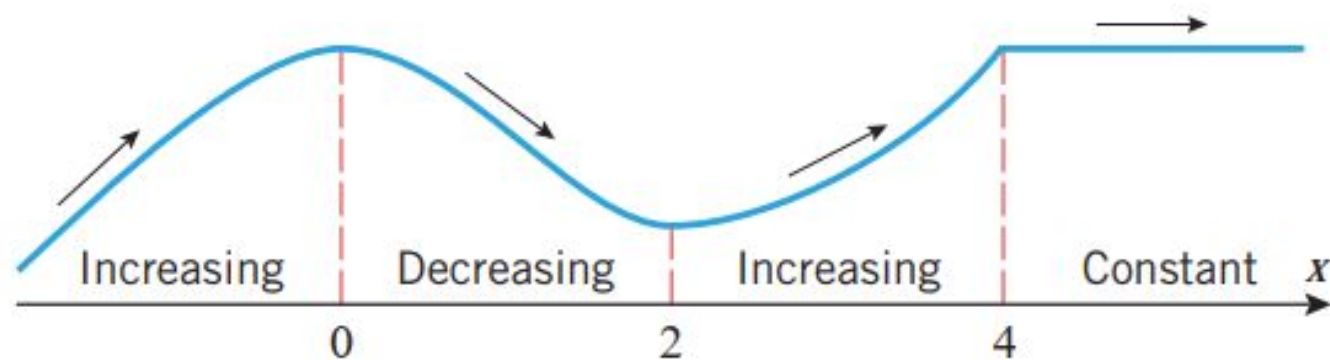
sign changes (between $0 < x < 2$ and $x > 2$)

→ The sign of f' changes from +ve to -ve at $x=0$, so there is a relative maximum at that point

→ The sign of f' changes from -ve to +ve at $x=2$, so there is a relative minimum at that point

■ INCREASING AND DECREASING FUNCTIONS

The terms *increasing*, *decreasing*, and *constant* are used to describe the behavior of a function as we travel left to right along its graph. For example, the function graphed in Figure 4.1.1 can be described as increasing to the left of $x = 0$, decreasing from $x = 0$ to $x = 2$, increasing from $x = 2$ to $x = 4$, and constant to the right of $x = 4$.



► Figure 4.1.1



2ND DERIVATIVE TEST

4.2.4 THEOREM (*Second Derivative Test*) Suppose that f is twice differentiable at the point x_0 .

- (a) If $f'(x_0) = 0$ and $f''(x_0) > 0$, then f has a relative minimum at x_0 .
- (b) If $f'(x_0) = 0$ and $f''(x_0) < 0$, then f has a relative maximum at x_0 .
- (c) If $f'(x_0) = 0$ and $f''(x_0) = 0$, then the test is inconclusive; that is, f may have a relative maximum, a relative minimum, or neither at x_0 .



► **Example 5** Find the relative extrema of $f(x) = 3x^5 - 5x^3$.

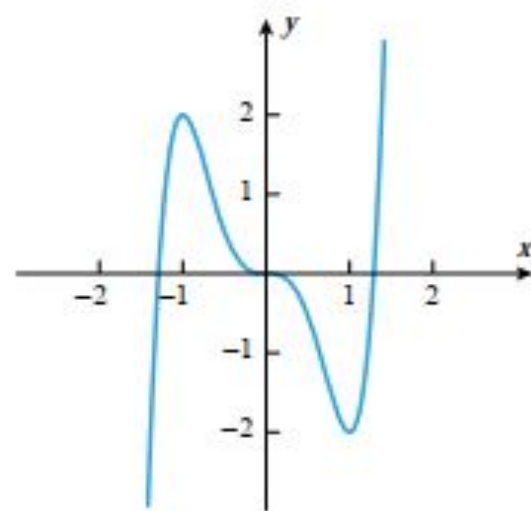
Solution. We have

$$f'(x) = 15x^4 - 15x^2 = 15x^2(x^2 - 1) = 15x^2(x + 1)(x - 1)$$

$$f''(x) = 60x^3 - 30x = 30x(2x^2 - 1)$$

Solving $f'(x) = 0$ yields the stationary points $x = 0$, $x = -1$, and $x = 1$. As shown in the following table, we can conclude from the second derivative test that f has a relative maximum at $x = -1$ and a relative minimum at $x = 1$.

STATIONARY POINT	$30x(2x^2 - 1)$	$f''(x)$	SECOND DERIVATIVE TEST
$x = -1$	-30	-	f has a relative maximum
$x = 0$	0	0	Inconclusive
$x = 1$	30	+	f has a relative minimum



$$y = 3x^5 - 5x^3$$

▲ Figure 4.2.8

The test is inconclusive at $x = 0$, so we will try the first derivative test at that point. A sign analysis of f' is given in the following table:

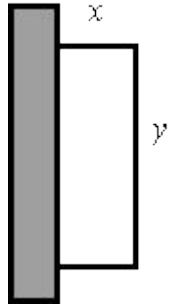
INTERVAL	$15x^2(x + 1)(x - 1)$	$f'(x)$
$-1 < x < 0$	(+)(+)(-)	-
$0 < x < 1$	(+)(+)(-)	-

Since there is no sign change in f' at $x = 0$, there is neither a relative maximum nor a relative minimum at that point. All of this is consistent with the graph of f shown in Figure 4.2.8. ◀



OPTIMIZATION

A rectangular storage area is to be constructed along the side of a tall building. A security fence is required along the remaining 3 sides of the area. What is the maximum area that can be enclosed with 800 m of fencing?



The area is $A = xy$

We know $2x + y = 800$ so $y = 800 - 2x$

So the area is $A = x(800 - 2x) = 800x - 2x^2$

To maximise the area, find when $\frac{dA}{dx} = 0$

$$\frac{dA}{dx} = 800 - 4x = 0$$

when

$$x = 200$$

Is it a maximum?

$$\frac{d^2A}{dx^2} = -4 < 0 \text{ for all } x$$

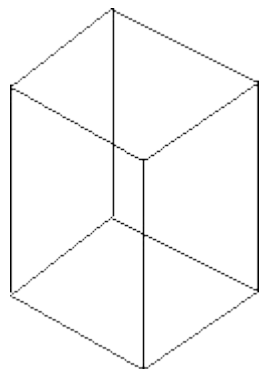
So it is a maximum.

So the maximum area occurs when $x = 200$, $y = 400$ and that area is:

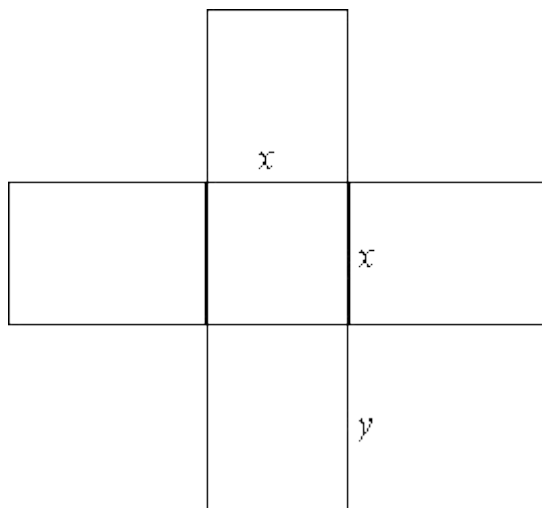
$$A = 200 \times 400 = 80\,000 \text{ m}^2 = 8 \text{ ha}$$



A box with a square base has no top. If 64 cm^2 of material is used, what is the maximum possible volume for the box?



The net for this box would be:



The volume of the box is $V = x^2y$

We are told that the surface area of the box is 64 cm^2 . The area of the base of the box is x^2 and the area of each side is xy , so the area of the base plus the area of the 4 sides is given by:

$$x^2 + 4xy = 64 \text{ cm}^2$$

Solving for y gives:

$$y = \frac{64 - x^2}{4x} = \frac{16}{x} - \frac{x}{4}$$

So the volume can be rewritten:

$$\begin{aligned} V &= x^2y \\ &= x^2 \left(\frac{16}{x} - \frac{x}{4} \right) \\ &= 16x - \frac{x^3}{4} \end{aligned}$$

Now

$$\frac{dV}{dx} = 16 - \frac{3x^2}{4}$$

and this is zero when

$$x = \pm \frac{8}{\sqrt{3}} \approx 4.62$$

(Note: The negative case has no practical meaning.)

Is it a maximum?

$$\frac{d^2V}{dx^2} = -\frac{3x}{2}$$

and this is negative when x is positive. So it is a MAX.

So the dimensions of the box are:

Base $4.62 \text{ cm} \times 4.62 \text{ cm}$ and sides 2.31 cm .

The maximum possible volume is

$$V = 4.62 \times 4.62 \times 2.31 \approx 49.3 \text{ cm}^3$$

Check: Area of material:

$$x^2 + 4xy = 21.3 + 4 \times 4.62 \times 2.31 = 64$$

Checks OK.



6.

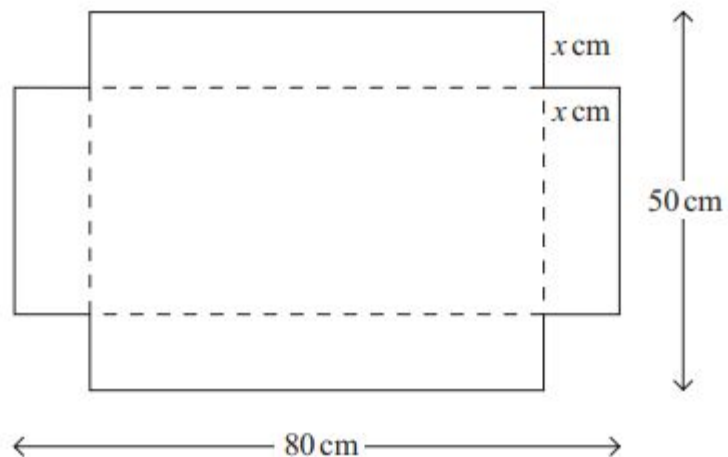


Figure 1

A rectangular sheet of card measures 80 cm by 50 cm. A square of side x cm is cut away from each corner, as shown in Figure 1. The card is folded along the dotted lines to form an open rectangular box of volume $V \text{ cm}^3$.

- (a) Show that $V = 4000x - 260x^2 + 4x^3$. (2)
- (b) Find the value of x for which V has its maximum value, justifying that this value of x gives the maximum value of V . (6)
- (c) Find the maximum value of V . (2)

